HEILONGJIANG WANGYUNFENG WIND POWER PROJECT

Document Prepared by Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.

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CONTENTS

1 PROJECT DETAILS 4
1.1 Summary Description of the Implementation Status of the Project 4
1.2 Sectoral Scope and Project Type 4
1.3 Project Proponent 4
1.4 Other Entities Involved in the Project 5
1.5 Project Start Date 5
1.6 Project Crediting Period 5
1.7 Project Location 5
1.8 Title and Reference of Methodology 7
1.9 Participation under other GHG Programs 7
1.10 Other Forms of Credit 7
1.11 Sustainable Development Contributions 8
2 SAFEGUARDS 10
2.1 No Net Harm 10
2.2 Local Stakeholder Consultation 10
2.3 AFOLU-Specific Safeguards 11
3 IMPLEMENTATION STATUS 11
3.1 Implementation Status of the Project Activity 11
3.2 Deviations 12
3.3 Grouped Projects 12
4 DATA AND PARAMETERS 12
4.1 Data and Parameters Available at Validation 12
4.2 Data and Parameters Monitored 13
4.3 Monitoring Plan 15
5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS 21
5.1 Baseline Emissions 21
5.2 Project Emissions 24
5.3 Leakage 24
5.4 Net GHG Emission Reductions and Removals 24

APPENDIX1: <SUPPORTING EVIDENCE> 25

1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Heilongjiang Wangyunfeng Wind Power Project (hereinafter refers to the project) is developed by Hegang Longyuan Wind Power Co., Ltd. and located on Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province, P.R.China. Totally 58 sets of 850kW wind turbines were installed, providing a total capacity of 49.3MW. It is estimated the power supply was 120,069 MWh annually and achieved amount of 123,430 tCO_2e emission reduction every year as per registered PD. The objective of the project is to generate electricity using state-of-the-art wind power generation technology and to sell into Northeast China Power Grid (NECPG).

Construction start permission of the project was issued on 04-Jun-2009 and the construction contract was signed on 05-Jun-2009. The first wind turbine started to commercial operation on 20-Jan-2010 and all wind turbines started commercial operation on 12-Apr-2010.

This is the 3rd VCS monitoring period from 01-Jan-2019 to 19-Jan-2020 under VCS program. During this monitoring period, the operation condition of all the wind turbines and the operation environment haven't been changed from the previous monitoring periods, which are also consistent with the description of the registered PD, the project has generated the total net electricity supply of 117,718.920 MWh and achieved 121,014 tCO_2e emission reduction.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 energy industries (Renewable sources).

Project type: Grid-connected wind power project.

The project is not a grouped project..

1.3 Project Proponent

Organization name	**Hegang Longyuan Wind Power Co., Ltd.**
Contact person	Mr. Zhang Yulei
Title	Marketing Manager
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Telephone	+86 451 51919150
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1.4 Other Entities Involved in the Project

Organization name	**Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.**
Role in the Project	Consultancy
Contact person	Shanshan Yu
Title	General Manager
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1.5 Project Start Date

20-Jan-2010 (the first wind turbine started to commercial operation).

1.6 Project Crediting Period

The first 10-year crediting period of the project under VCS is from 20-Jan-2010 to 19-Jan-2020 (both days included).

1.7 Project Location

The project is located in Damahe Forest Farm, Luobei County, Hegang City, Heilongjiang Province of China. Its central geographical coordinates are north latitude 47°54′34″, and east longitude 130°39′56″, its altitude is about 240m~430m. The detailed location of the project is shown in the Figure 1-1 below.

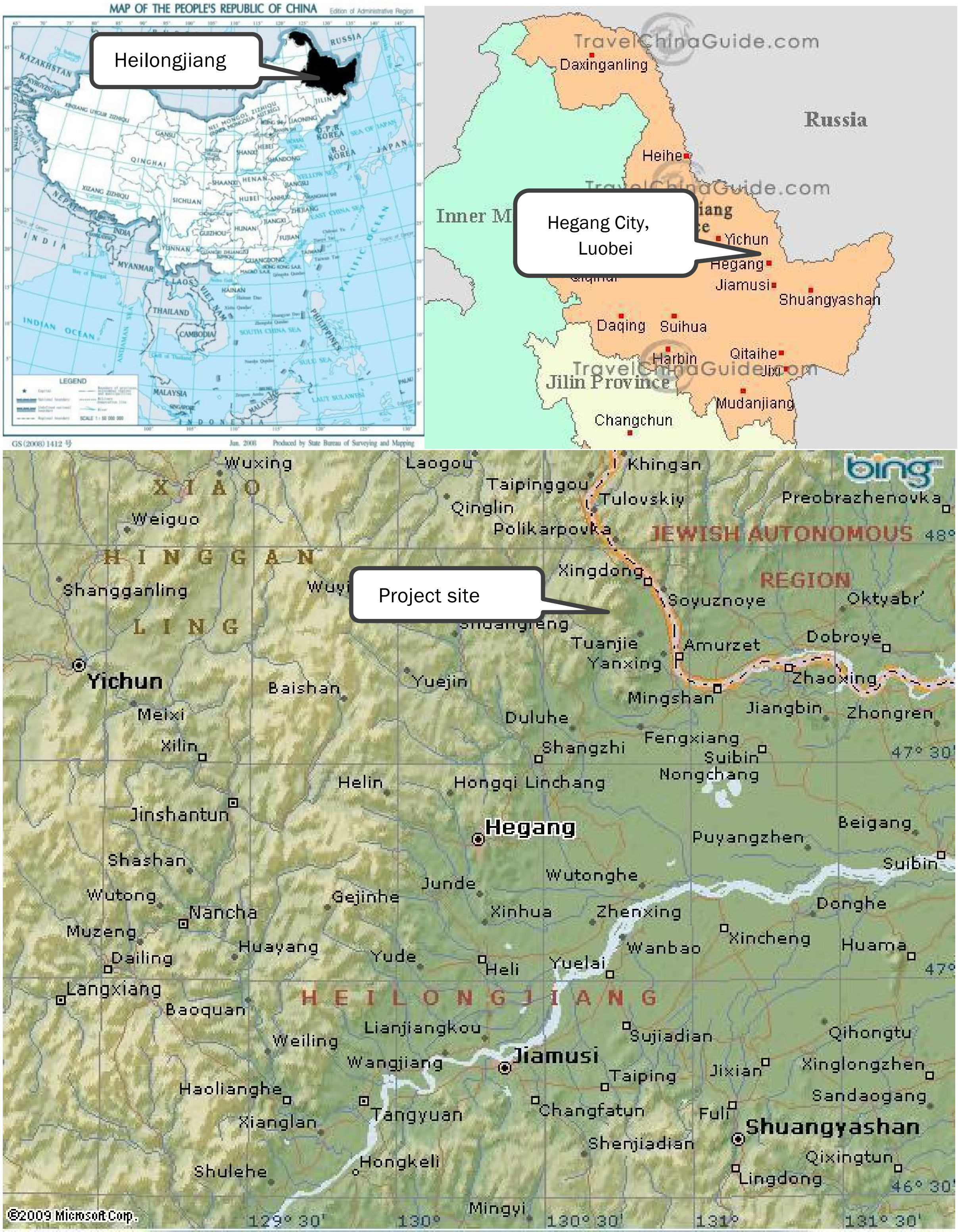


Figure 1-1 Location of the project

1.8 Title and Reference of Methodology

Title: Approved consolidated baseline and monitoring methodology ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources" (Version 12.1.0)

The methodology also refers to the following tools:

- Tool to calculate the emission factor for an electricity system (Version 02.2.1);
- Tool for the demonstration and assessment of additionality (Version 05.2.1).

https://cdm.unfccc.int/methodologies/PAmethodologies/approved

1.9 Participation under other GHG Programs

The project was registered as a CDM project on 31-Jan-2012 under the UNFCCC reference number 5711 with a first 7-year CDM renewable crediting period from 31-Jan-2012 to 30-Jan-2019, the detailed information can be found at https://cdm.unfccc.int/Projects/DB/BVQI1327230197.63/view

Start Date	End Date	GHG Emission Reduction (tCO_2e)	Credit	Program
20-Jan-2010	30-Jan-2012	195,043	VCU	VCS
31-Jan-2012	31-Dec-2012	72,155	CER	CDM
01-Jan-2013	31-Dec-2018	517,149	VCU	VCS

This is the third monitoring period under VCS program from 01-Jan-2019 to 19-Jan-2020 for the project. All emission reductions during this monitoring period have not and will not seek CDM CER issuance. Emission reductions during this monitoring period will only seek issuance under VCS program.

1.10 Other Forms of Credit

Emission Trading Programs and Other Binding Limits

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO_2e/year, not including renewable project[1].

[1] http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html

The project proponent Hegang Longyuan Wind Power Co., Ltd., as an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Also, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China's ETS. Therefore, the project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The net GHG emission reductions generated during this monitoring period have not been used for compliance under such programs or mechanisms.

Furthermore, as a voluntary emission reduction project, the GHG emission reductions and removals generated by the project will not be enforced to be used for compliance in China. A statement on no double counting will be submitted to Verra to confirm the credits during this monitoring period has not been counted and will not be counted under emission trading programs and other binding limits.

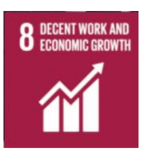
Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

1.11 Sustainable Development Contributions

The project activity is a renewable wind power project located in Heilongjiang Province, Northeast China. During this monitoring period,

The project utilizes wind resources to generate and supply 117,718.920 MWh renewable electricity to NECPG, which contributes to SDG 7.



The project provides 12 long-term job opportunities for local, which has a positive effect on the local economy and contributes to SDG 8.

The project reduces 121,014 tCO_2e GHG emission and contributes to SDG 13.

These contribute to achieve China's stated sustainable development priorities, included strengthening resistance capacity to climate risks, increasing labor force participation rate, increasing the share of clean energy consumption.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	Renewable energy supplied to the power grid	Implemented activities to increase	117,718.920 MWh of electricity from renewable sources has been supplied to the power grid during the reporting period.	An accumulated 810,518.212 MWh of electricity from renewable sources has been supplied to the power grid by end of the reporting period.
2)	8.5	Numbers of job opportunities provided by implementation of the project	Implemented activities to increase	12 people have been employed by the project as long-term employee whose salary higher above the national average line level.	12 people have been employed by the project as long-term employee whose salary higher above the national average line level by the end of the reporting period.
3)	13.0	Tonnes of greenhouse gas emissions avoided	Implemented activities to increase	The project has achieved a total GHG emission reduction of 121,014 $tCO2e$ during the reporting period.	An accumulated 833,206 tCO_2e of GHG emission reduction have been achieved by the project by end of the reporting period.

2 SAFEGUARDS

2.1 No Net Harm

Environmental Impact

The Environmental Impact Assessment (EIA) of the project has been conducted and approved by local government officials. The EIA report has assessed every possible aspect of environmental impacts of the project and recommended corresponding measures, where applicable. The environmental impacts and corresponding mitigation measures during operation have been discussed in the registered PD. No negative environmental impacts haven been identified.

Socio-economic impact

Furthermore, the project is renewable project and achieves the SDG 7 and SDG 13. Also, the project provides 12 long-term job opportunities for local, which has a positive effect on the local economy and contributes to SDG 8. Please refer to section 1.11 of this report for details.

In conclusion, no net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

LSC prior to the project implementation

Local stakeholder consultation has been conducted on 06-May-2009. All comments received have been well addressed prior to the project implementation. Please refer to section E of the registered PD (version 3 dated 21-Dec-2012) of the project for detailed information.

In conclusion, there was no significant negative comments received in the project preparation stage.

LSC during the operation period

Communications with Local stakeholders have been carried out at periodic intervals. Key implementation schedules and changes of the project have been communicated to the local authority, the neighbourhood committee, and local residents. The comments and suggestions from residents were collected by the local authority, if any.

The project owner carried out questionnaire survey for the local stakeholders to collect the relevant comments and suggestions every two years, e.g., in May 2019 during this monitoring period. The local authority also conducts spot checks on the implementation of the project at periodic intervals as per relevant regulations. There have been no negative comments received for the project so far.

In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

The project is a non-AFOLU project, this section is not required.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

Totally 58 sets of wind turbines with a total capacity of 49.3MW were installed and supply an average annual generation of 120,069MWh to NECPG. All wind turbines are produced in China. Technical Characteristics of Wind Turbines is shown as below table 3-1.

Each wind turbine has a box transformer which transforms the voltage to 35kV from which the 35kV lines link into the on-site 110kV switchgear at the project site. By the 110kV line, the electricity generated by the project is delivered to the power grid. The wind turbines and transmission facility could be monitored and controlled by onsite central control room. The technologies employed in the project activity are advanced domestic technologies, which is no technology transfer activity involved. The main advantage of this domestic manufactured turbine is its' adaptability to different types of wind resources and its improved generation efficiency, so it has been widely installed in China.

A technology diagram of the project is presented as Figure 4-2 in section 4.3.

During this monitoring period, the project was implemented in accordance with the registered PD. The project has operated without any accidental or emergency events that might impact the GHG emission reductions or removals and monitoring.

Main technical information of wind turbines used in the project

Table 3-1 Technical Characteristics of Wind Turbines[2]

Parameters	Data	
Rated wind speed（m/s）	13	
Unit capacity	850kW	
Type	G58-850	G52-850
Diameter of rotor（m）	58	52

[2] Sourced from Wind Turbines Purchase Agreement

Cut-in wind speed （m/s）	3
Cut-out wind speed （m/s）	21
Survival wind speed （m/s）	37.5
Swept area (m^2)	2642/2214
Lifetime	20 year
Generator	Doubly fed asynchronous generator

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology deviation in this monitoring period.

3.2.2 Project Description Deviations

As per registered VCS PD /3/, the VCS crediting period was determined as from 20-Jan-2010 to 30-Jan-2012. There is a deviation for the crediting period. The project is registered under VCS 3.2 and completed validation before 19-Mar-2020, thus it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project shall be updated from 20-Jan-2010~30-Jan-2012 to 20-Jan-2010~19-Jan-2020. This monitoring period belongs to the first crediting period from 20-Jan-2010~19-Jan-2020 (10 years). The project did not complete the validation of the crediting period update within two years of the end of the first crediting period, therefore the VCS total crediting period ends on 19-Jan-2020, the end of the first crediting period.

The deviation does not impact the applicability of the methodology, additionality and the appropriateness of the baseline scenario.

3.3 Grouped Projects

Not applicable.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF_y
Data unit	tCO_2/MWh
Description	Combined margin CO_2 emission factor for grid connected power generation in the year y

Source of data	Registered PD
Value applied	1.0280
Justification of choice of data or description of measurement methods and procedures applied	Ex-ante determined in the registered PD and fixed during the 1st crediting period.
Purpose of Data	Calculation of baseline emissions
Comments	Determined ex-ante and fixed for the 1st crediting period

4.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility, y}$
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Measured by electricity meters: M1
Description of measurement methods and procedures to be applied	The main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the backup meter M2 with accuracy of 0.5S installed at the inlet of the main transformer measure the quantity of the electricity exported to the grid by the project and the quantity of electricity imported from the grid by the project. Quantity of net electricity generation supplied by the project to the grid is the difference between the quantity of the electricity exported to the grid by the project and the quantity of electricity imported from the grid by the project.
Frequency of monitoring/recording	The reading of electricity meters are continuously measured, daily read and monthly aggregated.
Value monitored	Period: Value 01-Jan-2019~31-Dec-2019: 108,019.560 01-Jan-2020~19-Jan-2020: 9,699.360 Total: 117,718.920
Monitoring equipment	See as follow table 4-1
QA/QC procedures to be applied	The meters (bidirectional) are calibrated yearly according to national standards and rules.Double checked by receipt of sales. The designated person of the project reads and aggregates the number of the meter. Data is archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Purpose of the data	Calculation of baseline emissions
Calculation method	$EG_{facility,y} = EG_{export,y} - EG_{import,y}$

Value monitored:

Period	Value
01-Jan-2019~31-Dec-2019	108,019.560
01-Jan-2020~19-Jan-2020	9,699.360
Total	117,718.920

Comments	-

Data / Parameter	$EG_{export, y}$
Data unit	MWh
Description	Electricity exported to the grid by the project
Source of data	Measured by electricity meters: M1
Description of measurement methods and procedures to be applied	The main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the backup meter M2 with accuracy of 0.5S installed at the inlet of the main transformer measure the quantity of the electricity exported to the grid by the project and the quantity of electricity utilized by the project.
Frequency of monitoring/recording	The reading of electricity meters are continuously measured, daily read and monthly aggregated.
Value monitored	Period: 01-Jan-2019~31-Dec-2019 — Value: 108,258.480 Period: 01-Jan-2020~19-Jan-2020 — Value: 9,699.360 Total — Value: 117,957.840
Monitoring equipment	See as follow table 4-1
QA/QC procedures to be applied	The meters (bidirectional) are calibrated yearly according to national standards and rules.Double checked by receipt of sales. The designated person of the project reads and aggregates the number of the meter. Data is archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Purpose of the data	Calculation of baseline emissions
Calculation method	-
Comments	-

Value monitored:

Period	Value
01-Jan-2019~31-Dec-2019	108,258.480
01-Jan-2020~19-Jan-2020	9,699.360
Total	117,957.840

Data / Parameter	$EG_{import, y}$
Data unit	MWh
Description	Electricity utilized by the project
Source of data	Measured by electricity meters: M1
Description of measurement methods and procedures to be applied	The main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the backup meter M2 with accuracy of 0.5S installed at the inlet of the main transformer measure the quantity of the electricity exported to the grid by the project and the quantity of

	electricity utilized by the project.
Frequency of monitoring/recording	The reading of electricity meters are continuously measured, daily read and monthly aggregated.
Value monitored	<table><tr><td>Period</td><td>Value</td></tr><tr><td>01-Jan-2019~31-Dec-2019</td><td>238.920</td></tr><tr><td>01-Jan-2020~19-Jan-2020</td><td>0.000</td></tr><tr><td>Total</td><td>238.920</td></tr></table>
Monitoring equipment	See as follow table 4-1
QA/QC procedures to be applied	The meters (bidirectional) are calibrated yearly according to national standards and rules.Double checked by receipt of sales. The designated person of the project reads and aggregates the number of the meter. Data is archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Purpose of the data	Calculation of baseline emissions
Calculation method	-
Comments	-

4.3 Monitoring Plan

Introduction

The project adopts the approved consolidated baseline and monitoring methodology ACM0002 to determine the emission reductions from the wind farm. This plan describes in more detail the process.

Responsibility

Hegang Longyuan Wind Power Co., Ltd. is overall responsibility for monitoring and carries out the monitoring following this monitoring plan .The VCS manager of Hegang Longyuan Wind Power Co., Ltd. is responsible for the monitoring and reporting of the wind farm. The management structure is illustrated as follows:

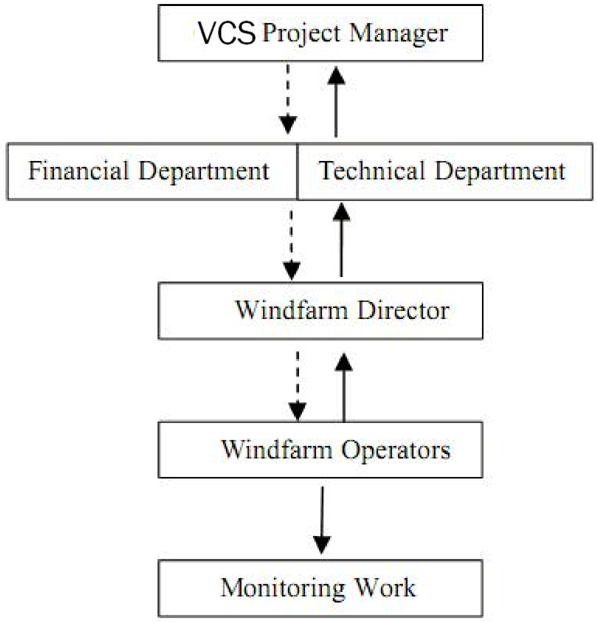


Figure 4-1 Organizational structure

Windfarm Director collects the information and data required by the Monitoring Plan. The collected information is documented and sent to the VCS manager and responsible staffs of Hegang Longyuan Wind Power Co., Ltd d monthly. The VCS manager is in charge of the implementation of the Monitoring Plan and the confirmations on monitoring, calculation data and report to the General Manager of the company. Managers of the project must maintain credible, transparent, and adequate data estimation, measurement, collection, and tracking systems to maintain the information required for an audit of an emission reduction project.

Monitoring equipment

Two bi-directional electricity meters are installed at the on-site 110 kV substation (the main meter M1 with accuracy of 0.5S installed at outlet of the main transformer and the meter M2 with accuracy of 0.5S installed at the inlet of the main transformer) to monitor directly $EG_{import,y}$ and $EG_{export,y}$ by the project (shown in the figure below). During the monitoring period, the main meter (M1) and the backup meter (M2) all worked normally. Therefore, the monitored data used in the MR were all sourced from the main meter (M1).

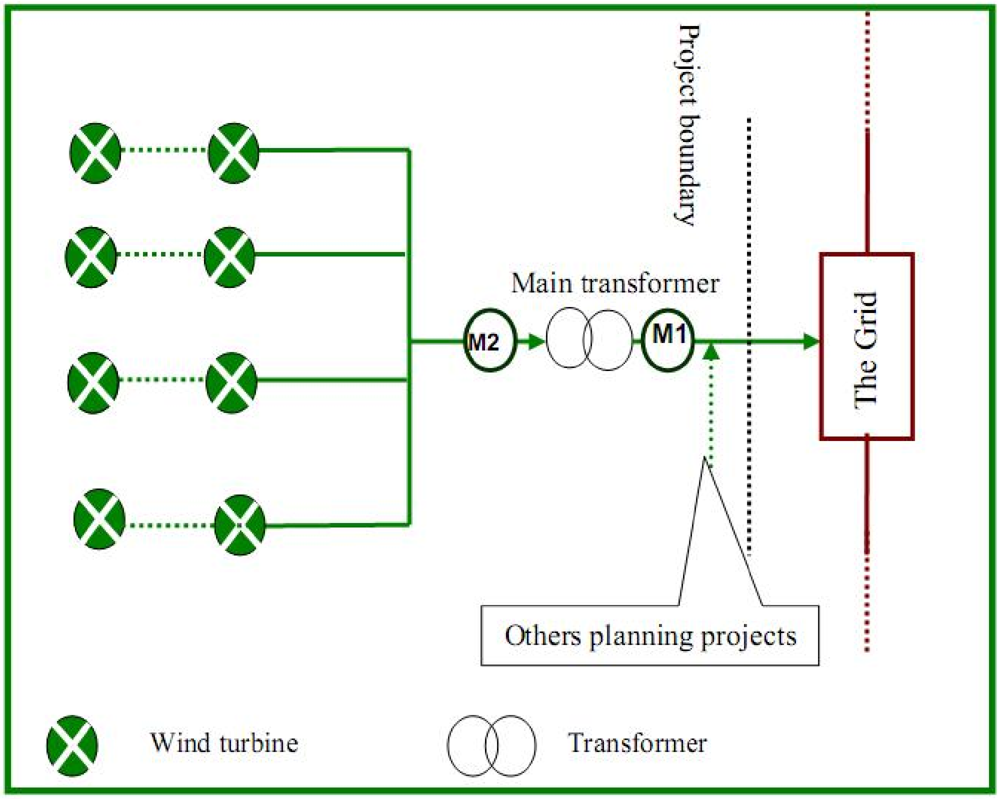


Figure 4-2 A technology diagram of the *project*

Calibration

The metering equipments related with the project are calibrated and tested yearly by a qualified third party appointed by the Northeast China Power Grid for accuracy according to the requirement from Technical administrative code of electric energy metering. The meters shall be jointly inspected and sealed on behalf of the parties concerned and shall not be interfered with by either party except in the presence of the other party or its accredited representatives.

Table 4-1 Monitoring Equipments of the Project

Meter	Type	Serial number	Calibrate Date	Validity date	Accuracy	Calibration Frequency	Calibration Entity
Main meter M1	DTSD719	4200525040909208867	30-Apr-2018 28-Apr-2019	29-Apr-2019 27-Apr-2020	0.5S	Annually	Hegang Metrological Verification Institution
Backup meter M2	DTSD719	4200222040909208865	30-Apr-2018 28-Apr-2019	29-Apr-2019 27-Apr-2020	0.5S	Annually	Hegang Metrological Verification Institution

Monitoring data

According to the illustration figure above, the net electricity supplied to the grid by the project is calculated by:

$EG_{facility,y} = EG_{export,y} - EG_{import,y}$

Data collection

The readings of all electricity meters are continuously measured, daily read and monthly aggregated. The designated staff collects the measured electricity data and complete the corresponding records on a monthly basis. The cut-off time of meter is at 24:00 of last day of the month.

Before being archived, these records are checked by other staffs to ensure the correctness. The data from these records is digested and analysed and the results are reported to company administrator or supervisor. Data will be archived for 2 years following the end of the last crediting period by means of electronic and paper backup.

Quality assurance and Quality control

The quality assurance and quality control procedures for the emergency about recording, maintaining and archiving data shall be improved as part of this VCS project activity. This is an on-going process that can be ensured through the VCS in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002.

Hegang Longyuan Wind Power Co., Ltd specially issued the regulations of monitoring and management for VCS project (hereafter refer as VCS Manual) to monitor and verify the emission reductions from the project. Moreover, the management procedures and regulations like operation, meter recordings, maintenance and emergency management are established in the VCS Manual, which guides the technicians to operate and guarantee the quality assurance and quality control procedures for recording, maintaining and archiving data in detail. This is an on-going process that can be ensured through the VCS in terms of the need for verification of the emissions on an annual basis according to methodology ACM0002.

Several processes required to meet the requirements for emissions reduction monitoring include:

The project owner reads the meters system monthly and supplies the recordings to the local Electric Power Company. The local Electric Power Company checks the recordings supplied by the project owner.

If either party finds any reading of the Meters more than the allowable error or the monitoring equipment functions improperly, it should inform the other party immediately. The project owner and the local Power Grid Company should retain a qualified measure institute together to check the meters or equipment, solve the problems and get everything into normal condition; In the mean time, both sides shall jointly prepare an acceptable estimation method according to Regulations and Rules of Power Supply1 issued by State Electricity Regulatory Commission of China to calculate the electricity exported to the Grid, and provide evidence to VVB for the verification to show the estimation is reasonable and conservative.

For the handling of disputes between the project owner and the Grid, measures are adopted according to relevant articles of Interim Measures for Settlement of Electricity Charges between the Power Generating Enterprises and the Grid Enterprises issued by State Electricity Regulatory Commission of China. Moreover, if meters are out of order, the electricity generation in the period is not considered.

Data Management System

Overall responsibility for monitoring greenhouse gas emissions reductions rests with the VCS monitoring staff of the project. The VCS manual sets out the procedures for tracking information from the primary source to data calculations in paper format. Moreover, the credibility and reliability of those data and information must be confirmed. Physical documentation such as paper-based maps, diagrams and environmental assessment are collated in a central place, together with this monitoring plan. In order to facilitate auditor's reference, monitoring results are indexed. All paper-based information will be stored by Hegang Longyuan Wind Power Co., Ltd and kept at least one copy, and all data including calibration records is kept until 2 years after the end of the total crediting period of the VCS project.

The following table below outlines the key documents relevant to monitoring and verification of the emission reductions from the project.

Table 4-2 The Key Documents of the Project

I.D.No.	Document Title	Main Content	Source
F-1	PD, including the electronic spreadsheets and supporting documentation(assumptions, estimations, measurement, etc)	Calculation procedure of emission reduction and monitoring items	Project owner or UNFCC website
F-2	Report on monitoring and checking of electricity supplied to the grid	Record based on monthly meter reading and electricity sale receipts	project owner
F-3	Report on maintenance and calibration of metering equipment	Reasons for maintenance and calibration and the precision after maintenance and calibration	project owner
F-4	Report on the qualifications of the operators	Technical post ,working experience etc.	project owner
F-5	the project management record (including date collection and management system)	Comprehensively and truly reflect the management and the operation of the project	project owner

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

According to ACM0002 and the registered PD of the Project, The baseline emission BE_y during the monitoring period results from:

$$BE_y = EG_{facility,\,y} \times EF_y = (EG_{export,\,y} - EG_{import,\,y}) \times EF_y \quad (1)$$

Where,

BE_y	Baseline emissions of the Project in year *y* (tCO_2e)
EF_y	Combined margin CO_2 emission factor for grid connected power generation (tCO_2e/MWh)
$EG_{facility,\,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)
$EG_{export,\,y}$	Electricity exported to the grid by the project (MWh)
$EG_{import,\,y}$	Electricity utilized by the project (MWh)

The monitored $EG_{facility,\,y}$ is reported in the table below.

Table 5-1 Monitored monthly value of $EG_{facility,y}$

	$EG_{export,y}$			$EG_{import,y}$			
Period	Monthly Recording from main meter (M1) (MWh)	Sales Receipt (MWh)	Conservative Quantity of the electricity exported to the grid (MWh)	Monthly Recording from main meter (M1) (MWh)	Purchase Receipt (MWh)	Conservative Quantity of the electricity imported from the grid (MWh)	$EG_{facility,y}$ (MWh)
	A	B	C=min(A,B)	D	E	F=max(D,E)	G=C-F
01/01/2019-31/01/2019	11,238.480	11,238.480	11,238.480	7.920	7.920	7.920	11,230.560
01/02/2019-28/02/2019	11,197.560	11,197.560	11,197.560	14.520	14.520	14.520	11,183.040
01/03/2019-31/03/2019	7,788.000	7,788.000	7,788.000	23.760	23.760	23.760	7,764.240
01/04/2019-30/04/2019	6,737.280	6,737.280	6,737.280	13.200	13.200	13.200	6,724.080
01/05/2019-31/05/2019	7,502.880	7,502.880	7,502.880	13.200	13.200	13.200	7,489.680
01/06/2019-30/06/2019	5,177.040	5,177.040	5,177.040	34.320	34.320	34.320	5,142.720
01/07/2019-31/07/2019	3,594.360	3,594.360	3,594.360	35.640	35.640	35.640	3,558.720
01/08/2019-31/08/2019	5,738.040	5,738.040	5,738.040	29.040	29.040	29.040	5,709.000
01/09/2019-30/09/2019	8,655.240	8,655.240	8,655.240	15.840	15.840	15.840	8,639.400
01/10/2019-31/10/2019	9,346.920	9,346.920	9,346.920	13.200	13.200	13.200	9,333.720
01/11/2019-30/11/2019	16,267.680	16,267.680	16,267.680	6.600	6.600	6.600	16,261.080
01/12/2019-31/12/2019	15,015.000	15,015.000	15,015.000	31.680	31.680	31.680	14,983.320
Subtotal	**108,258.480**	**108,258.480**	**108,258.480**	**238.920**	**238.920**	**238.920**	**108,019.560**
01/01/2020-19/01/2020	9,699.360	9,699.360	9,699.360	0.000	0.000	0.000	9,699.360
Subtotal	**9,699.360**	**9,699.360**	**9,699.360**	**0.000**	**0.000**	**0.000**	**9,699.360**
Total	**117,957.840**	**117,957.840**	**117,957.840**	**238.920**	**238.920**	**238.920**	**117,718.920**

The annual baseline emissions of the project are calculated as follows:

Table 5-2 Calculation of Baseline emissions

Year	$EG_{facility,y}$ (MWh)	EF_y (tCO_2/MWh)	BE_y (tCO_2)
	A	B	C=A*B
2019	108,019.560	1.0280	111,044
2020	9,699.360	1.0280	9,970
Total	**117,718.920**		**121,014**

In summary, the baseline emissions BE_y during this monitoring period is 121,014 tCO_2e.

5.2 Project Emissions

As per the applied methodology, the project emission is zero.

5.3 Leakage

As per the applied methodology, leakage of the project is not considered.

5.4 Net GHG Emission Reductions and Removals

This monitoring period started from 01-Jan-2019 to 19-Jan-2020, with totally 384 days. Based on the annual estimated emission reductions from the registered PD, the amount of emission reductions for this monitoring period would be 123,430 tCO_2e/365 d×384d=129,855 tCO_2e. The actual emission reductions in this monitoring period (384 days) are 121,014 tCO_2e, which are 6.81% lower than the estimation in the registered PD. It is because wind fluctuations and the limitation of the grid absorption capacity. So, it is reasonable that during this monitoring period, the actual emission reduction is less than the estimated value.

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
01-Jan-2019~31-Dec-2019	111,044	0	0	111,044
01-Jan-2020~19-Jan-2020	9,970	0	0	9,970
Total	**121,014**	**0**	**0**	**121,014**

APPENDIX1: <SUPPORTING EVIDENCE>

Evidence for SDG 7 and SDG 13: From project operation date onwards, the net electricity supplied to the grid and corresponding emission reduction in every monitoring period have been listed in below table. Please refer to following links for details.

https://registry.verra.org/app/projectDetail/VCS/1051

Period		**$EG_{facility,y}$**	**ER**	**Program**
From	To	MWh	tCO_2e	/
20-Jan-2010	30-Jan-2012	189,731.520	195,043	VCS
01-Jan-2013	31-Dec-2018	503,067.772	517,149	
01-Jan-2019	19-Jan-2020	117,718.920	121,014	
Total		810,518.21	833,206.00	

All impacts during the previous VCS monitoring period from 20-Jan-2010 to 31-Dec-2018 has been calculated in the cumulative impact. Thus, the cumulative impact has been calculated by summing the current project contributions with all impacts included in previously approved VCS monitoring reports or Sustainable Development Contribution Reports.

Evidence for SDG 8: The staff payroll of the project has been submitted as a separate document to VVB due to confidentiality requirement of the project proponent.